

Setup Oracle GoldenGate Veridata 12c

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Overview

1. IMPORTANT : Most of the scripts have hardcoded dbname, hostname , username and password.

a) 5 myDomain-ds.properties. files*

>>Change these lines based on the VIP, DB Name and VD prefix:

b) passwordfile.txt (1st line is for sys password)

c) create_domain_7001.py

>>Change these lines based on the domain path, hostname , weblogic user password , weblogic port

```
readTemplate('/u01/veridata/oradata/verirep/wls/wlserver/common/templates/wls/wls.jar') cmo.setPassword('welcome123') cmo.setListenPort(7001) cmo.setListenAddress('ggpsppf501.ie.intuit.net') writeDomain('/u01/veridata/oradata/verirep/wls/user_projects/domains/base_domain')
```

1. Preparation for Veridata Console

1.1. Create Goldengate Hub as Veridata server

```
./create_gg_hub_ha_PSP_PROD_UW2_Linux7_IDPS.sh psp prod pspvd1pt us-west-2 vpc-2 yes  
Change InstanceTypes for prod to m4.2xlarge in goldengate-hub-ha.json file as below:
```

```
"InstanceTypes": {  
  "GoldenGate": {  
    "qa": "m4.large",  
    "e2e": "m4.large",  
    "prf": "m4.large",  
    "stg": "m4.large",  
    "sbx": "m4.large",  
    "prod": "m4.xlarge"  
  }  
},  
to  
"InstanceTypes": {  
  "GoldenGate": {  
    "qa": "m4.large",  
    "e2e": "m4.large",  
    "prf": "m4.large",  
    "stg": "m4.large",  
    "sbx": "m4.large",  
    "prod": "m4.4xlarge"  
  }  
},
```

1.2. Create a RDS as Veridata repository db

`./create_rds.sh e2e epsvdpdt us-west-2 vpc-3`

Create the below Tablespaces for Veridata:

```
CREATE TABLESPACE VERIDATA DATAFILE SIZE 5G AUTOEXTEND ON NEXT 1G MAXSIZE UNLIMITED;  
CREATE TEMPORARY TABLESPACE VERIDATA_TEMP TEMPFILE SIZE 1G AUTOEXTEND ON NEXT 1G MAXSIZE UNLIMITED;
```

1.3. Setup the environment

#1. Ensure we have an Oracle DB setup for Veridata

#2. Required Dir path exists as per the need

```
mkdir -p /u01/veridata/wls  
mkdir -p /u01/veridata/Inv  
mkdir -p /u01/veridata/jdk1.8  
mkdir /tmp/setup_veridata_srv
```

1.4. Copy scripts and software

#1. Copy the below listed tar and sql files from S3 to Golden Gate Hub (Veridata Console) /tmp directory. You can also copy the files from git repo: https://github.com/qbo/qbo-ops-script-refreshed/tree/master/newdb/setup_veridata_srv

```
setup_veridata_srv.tar
veridata_tables.sql
veridata_procedures.sql
```

For e.g:

```
e2e: aws s3 cp s3://bits-sbg-qbo-ppd-us-west-2/binaries/goldengate/temp/Veridata/setup_veridata_srv.tar /tmp
/setup_veridata_srv/setup_veridata_srv.tar
\\
```

Prod:

```
mkdir /tmp/setup_veridata_srv
cd /tmp/setup_veridata_srv
aws s3 cp s3://bits-sbg-qbo-prod-us-west-2/binaries/goldengate/Veridata/setup_veridata_srv.tar .
\\
cd /u01/ogg/scripts
aws s3 cp s3://bits-sbg-qbo-prod-us-west-2/binaries/goldengate/Veridata/veridata_srv_scripts.tar .
\\
```

#2. Untar the files on the new Veridata Console server.

```
cd /tmp/setup_veridata_srv
tar -xvf setup_veridata_srv.tar
```

#3. On the new Veridata server, you should have the following files in the temp dir-

```
[oracle@ggppspf501 setup_veridata_srv]$ ls -ltr
total 4728360
-r-xr-xr-x. 1 oracle oinstall 1563217124 Oct  9  2016 fmw_12.2.1.2.0_infrastructure.jar
-rw-r--r--. 1 oracle oinstall      8986 Oct 11  2016 fmw_12212_readme.htm
-r-xr-xr-x. 1 oracle oinstall  446021430 Oct 11  2016 fmw_12.2.1.2.0_ogg.jar
-rw-----. 1 oracle oinstall  445820174 May 27 12:57 fmw_12.2.1.2.0_ogg_Disk1_1of1.zip
-rw-----. 1 oracle oinstall 1563100157 May 27 12:57 FWM_122120_infra_V779122-01.zip
-rw-----. 1 oracle oinstall  445820174 May 27 12:57 V789398-01.zip
-rw-----. 1 oracle oinstall    410 May 27 13:07 create_domain_7001.py
-rw-----. 1 oracle oinstall    337 May 27 13:07 enable_domain_7001.py
-rw-----. 1 oracle oinstall    342 May 27 13:07 enable_domain_7001_1.py
-rw-----. 1 oracle oinstall    338 May 27 13:07 enable_domain_7001_2.py
-rw-----. 1 oracle oinstall  377784320 May 27 13:07 jdk-1.8-u131-x64.tar
-rw-----. 1 oracle oinstall    428 May 27 13:07 myDomain-ds.properties.opsaudDBDS
-rw-----. 1 oracle oinstall    428 May 27 13:07 myDomain-ds.properties.opsaudviewDS
-rw-----. 1 oracle oinstall    416 May 27 13:07 myDomain-ds.properties.opss
-rw-----. 1 oracle oinstall    428 May 27 13:07 myDomain-ds.properties.stb
-rw-----. 1 oracle oinstall    421 May 27 13:07 myDomain-ds.properties.veridata
-rw-----. 1 oracle oinstall    61 May 27 13:07 oraInst.loc
-rw-----. 1 oracle oinstall    75 May 27 13:07 passwordfile.txt
-rw-----. 1 oracle oinstall   3013 May 27 13:16 create_data_source.py
-rwx-----. 1 oracle oinstall    599 May 28 11:31 veridata_srv_resp
-rwx-----. 1 oracle oinstall   1310 May 28 11:36 fusion_middleware_resp
```

#4. Correct Golden Gate Hub host name and RDS Repository instance connect string

```
create_domain_7001.py
myDomain-*
myDomain-*
```

#5. Make sure the below 2 lines are commented out in create_data_source.py:

```
\\
cd('/SystemResources/' + dsName)
set('Targets',jarray.array(\[ObjectName('com.bea:Name=' + dsTargetName + ',Type=' + dsTargetType)\],
ObjectName))
```

#6. Update the '<password>' & <wl-password> in all myDomain-* files (weblogic and Veridata users)

```
admin.password=<wl-password>
ds.password=<password>
```

2. Installation

2.1 Setup JDK

```
cd /tmp/setup_veridata_srv
unzip FWM_122120_infra_V779122-01.zip
unzip fmw_12.2.1.2.0_ogg_Disk1_1of1.zip
cp jdk-1.8-u131-x64.tar.gz /u01/veridata/jdk1.8/
/u01/veridata/jdk1.8/
tar -xvf /tmp/setup_veridata_srv/jdk-1.8-u131-x64.tar

update /l/<SID> file to add JAVA_HOME environment variable and add it to PATH and LD_LIBRARY_PATH variable

export JAVA_HOME=/u01/veridata/jdk1.8/jdk1.8.0_131
export PATH=${JAVA_HOME}/bin:${ORACLE_HOME}/bin:${ORACLE_HOME}:${PATH}:/l:/usr/sbin

export
LD_LIBRARY_PATH=${JAVA_HOME}/jre/lib/amd64/server:${ORACLE_HOME}/lib32:${ORACLE_HOME}/lib:${ORACLE_HOME}/jdbc
/lib:/usr/openwin/lib:/usr/dt/lib

Source DB profile file
. /l/<SID> or . /l/orcl (aws)
```

2.2 Setup Weblogic Server

```

export temp_dir=/tmp/setup_veridata_srv
export MW_HOME=/u01/veridata/wls
export WLS_HOME=$MW_HOME/wlserver
cd $temp_dir
#export _JAVA_OPTIONS="-Djava.io.tmpdir=$temp_dir"

java -jar fmw_12.2.1.2.0_infrastructure.jar -silent -responseFile $temp_dir/fusion_middleware_resp -invPtrLoc
$temp_dir/oraInst.loc

##### Output should be like below on completion #####

[oracle@ggppsppf501 setup_veridata_srv]$ java -jar fmw_12.2.1.2.0_infrastructure.jar -silent -responseFile
$temp_dir/fusion_middleware_resp -invPtrLoc $temp_dir/oraInst.loc
Launcher log file is /tmp/OraInstall2020-05-28_11-39-03AM/launcher2020-05-28_11-39-03AM.log.
Extracting the installer . . . . . Done
Checking if CPU speed is above 300 MHz.    Actual 2300.070 MHz    Passed
Checking swap space: must be greater than 512 MB.    Actual 1023 MB    Passed
Checking if this platform requires a 64-bit JVM.    Actual 64    Passed (64-bit not required)
Checking temp space: must be greater than 300 MB.    Actual 29766 MB    Passed

Preparing to launch the Oracle Universal Installer from /tmp/OraInstall2020-05-28_11-39-03AM
Log: /tmp/OraInstall2020-05-28_11-39-03AM/install2020-05-28_11-39-03AM.log
Copyright (c) 1996, 2016, Oracle and/or its affiliates. All rights reserved.
Reading response file..
Skipping Software Updates
Starting check : CertifiedVersions
Expected result: One of oracle-6,oracle-7,redhat-7,redhat-6,SuSE-11,SuSE-12
Actual Result: redhat-7.4
Check complete. The overall result of this check is: Passed
CertifiedVersions Check: Success.

Starting check : CheckJDKVersion
Expected result: 1.8.0_101
Actual Result: 1.8.0_131
Check complete. The overall result of this check is: Passed
CheckJDKVersion Check: Success.

Validations are enabled for this session.
Verifying data
Copying Files
Percent Complete : 10
Percent Complete : 20
Percent Complete : 30
Percent Complete : 40
Percent Complete : 50
Percent Complete : 60
Percent Complete : 70
Percent Complete : 80
Percent Complete : 90
Visit http://www.oracle.com/support/policies.html for Oracle Technical Support policies.
Percent Complete : 100

The installation of Oracle Fusion Middleware 12c Infrastructure 12.2.1.2.0 completed successfully.
Logs successfully copied to /u01/veridata/oraInventory/logs.
#####

```

2.3 Setup Oracle GoldenGate Veridata

```

export temp_dir=/tmp/setup_veridata_srv
export MW_HOME=/u01/veridata/wls
export WLS_HOME=$MW_HOME/wlserver
cd $temp_dir

# export _JAVA_OPTIONS="-Djava.io.tmpdir=$temp_dir"
java -jar fmw_12.2.1.2.0_ogg.jar -silent -responseFile $temp_dir/veridata_srv_resp -invPtrLoc $temp_dir/oraInst.loc

##### Output should be like below on completion #####
[oracle@ggppsppf501 setup_veridata_srv]$ java -jar fmw_12.2.1.2.0_ogg.jar -silent -responseFile $temp_dir/veridata_srv_resp -invPtrLoc $temp_dir/oraInst.loc
Picked up _JAVA_OPTIONS: -Djava.io.tmpdir=/tmp/setup_veridata_srv
Launcher log file is /tmp/setup_veridata_srv/OraInstall2020-05-28_11-53-47AM/launcher2020-05-28_11-53-47AM.log.
Extracting the installer . . . . Done
Checking if CPU speed is above 300 MHz.    Actual 2300.070 MHz    Passed
Checking swap space: must be greater than 512 MB.    Actual 1023 MB    Passed
Checking if this platform requires a 64-bit JVM.    Actual 64    Passed (64-bit not required)
Checking temp space: must be greater than 300 MB.    Actual 30813 MB    Passed

Preparing to launch the Oracle Universal Installer from /tmp/setup_veridata_srv/OraInstall2020-05-28_11-53-47AM
Picked up _JAVA_OPTIONS: -Djava.io.tmpdir=/tmp/setup_veridata_srv
Log: /tmp/setup_veridata_srv/OraInstall2020-05-28_11-53-47AM/install2020-05-28_11-53-47AM.log
Copyright (c) 2016, Oracle and/or its affiliates. All rights reserved.
Reading response file..
Skipping Software Updates
Starting check : CertifiedVersions
Expected result: One of oracle-6,oracle-7,redhat-7,redhat-6,SuSE-11,SuSE-12
Actual Result: redhat-7.4
Check complete. The overall result of this check is: Passed
CertifiedVersions Check: Success.

Starting check : CheckJDKVersion
Expected result: 1.8.0_101
Actual Result: 1.8.0_131
Check complete. The overall result of this check is: Passed
CheckJDKVersion Check: Success.

Validations are enabled for this session.
Verifying data
Copying Files
Percent Complete : 10
Percent Complete : 20
Percent Complete : 30
Percent Complete : 40
Percent Complete : 50
Percent Complete : 60
Percent Complete : 70
Percent Complete : 80
Percent Complete : 90
Percent Complete : 100

The installation of Oracle Fusion Middleware 12c GoldenGate Monitor & Veridata 12.2.1.2.0 completed successfully.
Logs successfully copied to /u01/veridata/oraInventory/logs.
#####

MANUAL WORK
Add the last third line in the listed sql file below the mentioned ALTER SESSION stmt

vi $MW_HOME/oracle_common/common/sql/opss/scripts/opss_tables.sql
ALTER SESSION SET CURRENT_SCHEMA=&&1 ;
alter session set NLS_LENGTH_SEMANTICS=BYTE;

```

2.4 Create Repository

NOTE: 1. "-schemaPrefix " will add the specified prefix to the db usernames that will be added as part of installation. For example by giving VD1, we will have VD1_VERIDATA user in the db.

2. \$temp_dir/passwordfile.txt file has 7 passwords: first is sys db user password; second is for weblogic user password; next 5 passwords are for the new users being created at this step. Please make sure that sys and weblogic users' passwords are correct.

3. If any of these below steps fails, rename wls directory under /u01/veridata/ and restart from step #2.2

```
cd $MW_HOME/oracle_common/bin
```

```
./rcu -silent -createRepository -databaseType ORACLE -connectString pspfib1.sbg-ppp-ppd.a.intuit.com:1521:
psppfib1 -dbUser intuadmin -schemaPrefix VD \
-component IAU -tablespace VERIDATA -tempTablespace VERIDATA_TEMP \
-component IAU_APPEND -tablespace VERIDATA -tempTablespace VERIDATA_TEMP \
-component IAU_VIEWER -tablespace VERIDATA -tempTablespace VERIDATA_TEMP \
-component OPSS -tablespace VERIDATA -tempTablespace VERIDATA_TEMP \
-component VERIDATA -tablespace VERIDATA -tempTablespace VERIDATA_TEMP \
-component STB -tablespace VERIDATA -tempTablespace VERIDATA_TEMP -f < $temp_dir/passwordfile.txt
```

Output should be like below on completion

```
[oracle@ggppsppf501 bin]$ ./rcu -silent -createRepository -databaseType ORACLE -connectString pspfib1.sbg-ppp-
ppd.a.intuit.com:1521:psppfib1 -dbUser intuadmin -schemaPrefix VD \
> -component IAU -tablespace VERIDATA -tempTablespace VERIDATA_TEMP \
> -component IAU_APPEND -tablespace VERIDATA -tempTablespace VERIDATA_TEMP \
> -component IAU_VIEWER -tablespace VERIDATA -tempTablespace VERIDATA_TEMP \
> -component OPSS -tablespace VERIDATA -tempTablespace VERIDATA_TEMP \
> -component VERIDATA -tablespace VERIDATA -tempTablespace VERIDATA_TEMP \
> -component STB -tablespace VERIDATA -tempTablespace VERIDATA_TEMP -f < $temp_dir/passwordfile.txt
Picked up _JAVA_OPTIONS: -Djava.io.tmpdir=/tmp/setup_veridata_srv
```

RCU Logfile: /tmp/setup_veridata_srv/RCU2020-05-28_12-07_87212572/logs/rcu.log

Processing command line

Repository Creation Utility - Checking Prerequisites

Checking Global Prerequisites

Repository Creation Utility - Checking Prerequisites

Checking Component Prerequisites

Repository Creation Utility - Creating Tablespaces

Validating and Creating Tablespaces

Repository Creation Utility - Create

Repository Create in progress.

Percent Complete: 13

Percent Complete: 32

Percent Complete: 32

Percent Complete: 34

Percent Complete: 36

Percent Complete: 38

Percent Complete: 38

Percent Complete: 38

Percent Complete: 47

Percent Complete: 47

Percent Complete: 58

Percent Complete: 58

Percent Complete: 58

Percent Complete: 66

Percent Complete: 66

Percent Complete: 77

Percent Complete: 77

```

Percent Complete: 77
Percent Complete: 85
Percent Complete: 85
Percent Complete: 89
Percent Complete: 89
Percent Complete: 99
Percent Complete: 100
log4j:WARN No appenders could be found for logger (org.apache.commons.jelly.JellyContext).
log4j:WARN Please initialize the log4j system properly.
log4j:WARN See http://logging.apache.org/log4j/1.2/faq.html#noconfig for more info.

Repository Creation Utility: Create - Completion Summary

Database details:
-----
Host Name                : psppfib1.sbg-ppp-a.intuit.com
Port                     : 1521
Service Name              : PSPPFIB1
Connected As              : intuadmin
Prefix for (prefixable) Schema Owners : VD
RCU Logfile               : /tmp/setup_veridata_srv/RCU2020-05-28_12-07_87212572/logs/rcu.log

Component schemas created:
-----
Component                Status      Logfile

Common Infrastructure Services Success    /tmp/setup_veridata_srv/RCU2020-05-28_12-07_87212572
/logs/stb.log
Oracle Platform Security Services Success    /tmp/setup_veridata_srv/RCU2020-05-28_12-07_87212572
/logs/opss.log
Veridata Repository      Success    /tmp/setup_veridata_srv/RCU2020-05-28_12-07_87212572
/logs/veridata.log
Audit Services           Success    /tmp/setup_veridata_srv/RCU2020-05-28_12-07_87212572
/logs/iau.log
Audit Services Append    Success    /tmp/setup_veridata_srv/RCU2020-05-28_12-07_87212572
/logs/iau_append.log
Audit Services Viewer    Success    /tmp/setup_veridata_srv/RCU2020-05-28_12-07_87212572
/logs/iau_viewer.log

Repository Creation Utility - Create : Operation Completed

#####

```

2.5 Configure Domain for Oracle GoldenGate Veridata

```

cd $MW_HOME/oracle_common/common/bin/
nohup ./wlst.sh $temp_dir/create_domain_7001.py &

cd $MW_HOME/oracle_common/common/bin/
nohup ./wlst.sh $temp_dir/enable_domain_7001_1.py &

cd $MW_HOME/user_projects/domains/base_domain/bin/
nohup ./startWebLogic.sh >/dev/null 2>&1 &

export DOMAIN_HOME=$MW_HOME/user_projects/domains/base_domain
. $DOMAIN_HOME/bin/setDomainEnv.sh

java weblogic.WLST $temp_dir/create_data_source.py -p $temp_dir/myDomain-ds.properties.opss
java weblogic.WLST $temp_dir/create_data_source.py -p $temp_dir/myDomain-ds.properties.opsaudviewDS
java weblogic.WLST $temp_dir/create_data_source.py -p $temp_dir/myDomain-ds.properties.opsaudDBDS
java weblogic.WLST $temp_dir/create_data_source.py -p $temp_dir/myDomain-ds.properties.veridata
java weblogic.WLST $temp_dir/create_data_source.py -p $temp_dir/myDomain-ds.properties.stb

##### Output would like similar to below with username changes as shown in
screenshot ###
[oracle@ggpppf501 base_domain]$ java weblogic.WLST $temp_dir/create_data_source.py -p $temp_dir/myDomain-ds.
properties.opss

```


Initializing WebLogic Scripting Tool (WLST) ...

Welcome to WebLogic Server Administration Scripting Shell

Type help() for help on available commands

```
properties= /tmp/setup_veridata_srv/myDomain-ds.properties.opss
adminUsername= weblogic
adminPassword= welcome123
adminURL= t3://ggpsppf501.sbg-ppsp-ppd.a.intuit.com:7001
dsName= opss-data-source
dsJNDIName= jdbc/OpssDataSource
dsURL= jdbc:oracle:thin:@psppfib1.sbg-ppsp-ppd.a.intuit.com:1521:psppfib1
dsDriver= oracle.jdbc.xa.client.OracleXADataSource
dsUsername= vd_opss
dsPassword= welcome123
dsTargetType= Cluster
dsTargetName= AdminServer
Connecting to t3://ggpsppf501.sbg-ppsp-ppd.a.intuit.com:7001 with userid weblogic ...
Successfully connected to Admin Server "AdminServer" that belongs to domain "base_domain".
```

Warning: An insecure protocol was used to connect to the server.
To ensure on-the-wire security, the SSL port or Admin port should be used instead.

Location changed to edit tree.
This is a writable tree with DomainMBean as the root.
To make changes you will need to start an edit session via startEdit().
For more help, use help('edit').

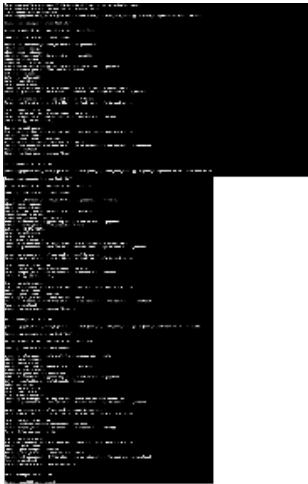
```
Starting an edit session ...
Started edit session, be sure to save and activate your changes once you are done.
Saving all your changes ...
Saved all your changes successfully.
Activating all your changes, this may take a while ...
The edit lock associated with this edit session is released once the activation is completed.
Activation completed
Disconnected from weblogic server: AdminServer
```

Exiting WebLogic Scripting Tool.

#####

```
export temp_dir=/tmp/setup_veridata_srv
export MW_HOME=/u01/veridata/wls
export WLS_HOME=$MW_HOME/wlserver
```

```
cd $MW_HOME/oracle_common/common/bin/
nohup ./wlst.sh $temp_dir/enable_domain_7001_2.py &
```



2.5a Setup up WebLogic ELB

- (one time only per environment) Download the script `create_veridata_security_group.sh` from the end of this Wiki
- (one time only per environment) Run the following command to create the security group for ELB
- Modify the veridata ELB security group by adding `<your IP>/32` on http (port 80) temporarily. You can get your IP by running "`curl {+}https://api.ipify.org+`" from your machine. You should delete this entry after complete the Veridata configuration.
- (one time only per environment) Add ports 7001 and 8830 with the source of the security group created above to the security group `prod-goldengate-vpc-3`
- Login to the AWS console and choose `us-west-2` (Oregon)
- Go to `EC2=>Load Balances`
- Click `Create Load Balance`
- Click `Create` under `Classic Load Balance`
- Enter "`prod-veridata-weblogic`" as Load Balancer name
- Select `Intuit-vpc-3` as Create LB Inside
- Enter 7001 as Instance Port
- Choose `IngressSubnetAz1` and `IngressSubnetAz2` subnets
- Unselect default security group and select `prod-veridata-vpc-3` security group
- Enter "`/console/login/LoginForm.jsp`" as Ping Path
- Choose Veridata server EC2 instance `goldengate-qbpvd1pt-xxx`
- Add Name as Key and "`prod-veridata-weblogic`" as value on Add Tags section
- Click `Review and Create`
- Click `Create`
- After Weblogic is started, go to `Ec2=>Load Balances` and search newly created load balance
- Click `Instances` to make sure that the status is "`inService`". Otherwise, you need to check whether Weblogic is started successfully
- Click `Description`, copy the DNS Name's value and then past `<dns name>/console/login/LoginForm.jsp` to browser. You will login to Weblogic.

2.5b Verify Data Sources

- Login to WebLogic Server and goto `base_domain->Services->Data Sources`. You should see the below 5 datasources
- WebLogic Server URL: `{+}http://e2e-goldengate-veridata-weblogic-836817812.us-west-2.elb.amazonaws.com/console+`

ORACLE WebLogic Server Administration Console 12c

Home Log Out Preferences Record Help Welcome, weblogic Connected to: base_dom

Home > myrealm > Users and Groups > Credential Mappings > Users and Groups > Summary of Services > Summary of Deployments > base_domain > Summary of Servers > Summary of Services: JMS > Summary of JDBC Data Sources

Summary of JDBC Data Sources

Configuration Monitoring

A JDBC data source is an object bound to the JNDI tree that provides database connectivity through a pool of JDBC connections. Applications can look up a data source on the JNDI tree and then borrow a database connection from a data source.

This page summarizes the JDBC data source objects that have been created in this domain.

[Customize this table](#)

Data Sources (Filtered - More Columns Exist)

Name	Type	JNDI Name	Targets
LocalSvcTblDataSource	Generic	jdbc/LocalSvcTblDataSource	AdminServer
opss-audit-DBDS	Generic	jdbc/AuditAppendDataSource	AdminServer, VERIDATA_server1
opss-audit-viewDS	Generic	jdbc/AuditViewDataSource	AdminServer, VERIDATA_server1
opss-data-source	Generic	jdbc/OpssDataSource	AdminServer, VERIDATA_server1
VeridataDataSource	Generic	VeridataDataSource	VERIDATA_server1

Showing 1 to 5 of 5 Previous Next

2.5c Create Veridata user data

Login to Weblogic Server UI (<http://pprdqbodb91b.ie.intuit.net:7001/>)
Security Realms --> myrealm --> User and Groups --> New

Create a New User

OK Cancel

User Properties

The following properties will be used to identify your new User.
* Indicates required fields

What would you like to name your new User?

* Name: vdata

How would you like to describe the new User?

Description: vdata

Please choose a provider for the user.

Provider: DefaultAuthenticator

The password is associated with the login name for the new User.

* Password:

* Confirm Password:

OK Cancel

This page displays information about each user that has been configured in this security realm.

Customize this table

Users (Filtered - More Columns Exist)

New Delete

Showing 1 to 3 of 3 Previous | Next

☐	Name ↕	Description	Provider
☐	OracleSystemUser	Oracle application software system user.	DefaultAuthenticator
☐	vdata	vdata	DefaultAuthenticator
☐	weblogic	This user is the default administrator.	DefaultAuthenticator

New Delete

Showing 1 to 3 of 3 Previous | Next

Click 'vdata'
Select 'Groups' Tab and choose Veridata* from Parents Groups and then Save

Settings for vdata

General

Passwords

Attributes

Groups

Save

Use this page to configure group membership for this user.

Parent Groups:

Available:

☐ AdminChannelUsers

☐ Administrators

☐ AppTesters

☐ CrossDomainConnectors

☐ Deployers

☐ Monitors

☐ Operators

☐ OracleSystemGroup

>

>>

<

<<

Chosen:

☐ veridataAdministrator

☐ veridataCommandLineUser

☐ veridataDetailReportViewer

☐ veridataPowerUser

☐ veridataRepairOperator

☐ veridataReportViewer

Save

2.6a Manual Settings

```

-- manual work
----- web logic changes -----
cd $MW_HOME/user_projects/domains/base_domain/bin
cp setStartupEnv.sh setStartupEnv.sh.orig
vi setStartupEnv.sh and look for # Startup parameters for STARTUP_GROUP VERIDATA-MANAGED-SERVERS
>>>under the STARTUP_GROUP VERIDATA-MANAGED-SERVERS change

SERVER_MEM_ARGS_64HotSpot="-Xms256m -Xmx512m -XX:PermSize=128m -XX:MaxPermSize=512m" To:
SERVER_MEM_ARGS_64HotSpot="-Xms32768m -Xmx32768m -XX:PermSize=2048m -XX:MaxPermSize=2048m"

chmod 777 setStartupEnv.sh

>>> Restart the weblogic server

cd $MW_HOME/user_projects/domains/base_domain/bin/
./stopWebLogic.sh
nohup ./startWebLogic.sh >/dev/null 2>&1 &

-----veridata changes -----
cd $MW_HOME/user_projects/domains/base_domain/config/veridata
cp veridata.cfg veridata.cfg.orig
vi veridata.cfg and edit below items
# SERVER PARAMETERS
server.meta_session_handle_timeout 900
server.max_concurrent_jobs 100
server.max_concurrent_comparison_threads default
server.max_sort_memory 17179869184
server.max_comparison_sort_memory 17179869184
server.sort_waiting_threshold 1000
server.concurrent.writers default
server.concurrent.readers default
server.number_sort_threads default

>>>Start the veridata server

cd $MW_HOME/user_projects/domains/base_domain/veridata/bin
./veridataServer.sh start

cd /u01/veridata/wls/user_projects/domains/base_domain/veridata/bin
./veridataServer.sh stop

cd $MW_HOME/user_projects/domains/base_domain/servers/VERIDATA_server1
mkdir security
cd security

>>>Create a text file named "boot.properties" with these two lines and save it
vi boot.properties
username=weblogic
password=welcome123

cd $MW_HOME/user_projects/domains/base_domain/veridata/bin
nohup ./veridataServer.sh start >/dev/null 2>&1 &

```

2.6b Setup up Veridata ELB

- Login to the AWS console and choose us-west-2 (Oregion)
- Go to EC2=>Load Balances
- Click Create Load Balance
- Click Create under Classic Load Balance
- Enter "prod-veridata" as Load Balancer name
- Select Intuit-vpc-3 as Create LB Inside
- Enter 8830 as Instance Port
- Choose IngressSubnetAz1 and IngressSubnetAz2 subnets
- Unselect default security group and select prod-veridata-vpc-3 security group
- Enter "/veridata/login.jsf" as Ping Path
- Choose Veridata server EC2 instance goldengate-qbpvd1pt-xxx
- Add Name as Key and "prod-veridata" as value on Add Tags section
- Click Review and Create

- Click Create
- After Veridata is started, go to Ec2=>Load Balances and search newly created load balance
- Click Instances to make sure that the status is "inService". Otherwise, you need to check whether Veridata is started successfully
- Click Description, copy the DNS Name's value and then past <dns name>/veridata/login.jsf to browser. You will login to Veridata

2.8 Setup Veridata profile sort_profile_new

In Veridata UI > Profile Configuration -> New

New Profile Assistant

Task 1: Profile Name

Please specify profile name and description

☐ Create new profile using copy of existing profile.

From:

Browse

Name

sort_profile_new

Description

sort_profile_new

Cancel

Finish

Save

Profile Details

Name:

sort_profile_new

Description:

Profile Settings

Category

- General
- Sorting Method
- Initial Compare
- Confirm-Out-Of-Sync
- Repair

General

Setting	Use Default	Value
Out-Of-Sync Output Format	☒	Binary
Maximum Size of Each Out-Of-Sync XML Chunk (Rows)	☒	500
Report in-sync rows to report file	☒	☐
Report in-sync after in-flight rows to report file	☒	☐

Save

Profile Settings			
Category - General - Sorting Method - Initial Compare - Confirm-Out-Of-Sync - Repair	Sorting Method		
	Setting	Use Default	Value
	Sort Data Using	☐	Server ▼
	Maximum Memory Usage (MB)	☐	1024 ◆
	Temporary Storage Directory for Source Data	☐	/u01/veridata/veri_s
	Temporary Storage Directory for Target Data	☐	/u01/veridata/veri_s

For Sorting Method,enter the below value for 'Temporary Storage Directory for Source Data' and 'Temporary Storage Directory for Target Data' :
/u01/veridata/veri_sort

Profile Settings			
Category - General - Sorting Method - Initial Compare - Confirm-Out-Of-Sync - Repair	Initial Compare		
	Setting	Use Default	Value
	Max Concurrent Comparison Threads	☐	10 ◆
	Terminate when Maximum Records Out-Of-Sync	☒	100000 ◆
	Output Out-Of-Sync Record Details to Report File	☐	☒
	Update Report file Every (seconds)	☒	0 ◆
	Update Report file Every (records)	☒	0 ◆
	Limit Number of Input Rows	☒	0 ◆
	Source Oracle optimizer hint	☒	
	Target Oracle optimizer hint	☒	
	Event Reporting		
	Setting	Use Default	Value
	Generate Messages	☒	None ▼
	Generate Warning Messages For Out-Of-Sync Rows After (differences)	☒	50 ◆
	Agent		
	Setting	Use Default	Value
	Use Static Listening Port For Agent During Row Hash On Source (0 to use dynamic port list)	☒	0 ◆
	Use Static Listening Port For Agent During Row Hash On Target (0 to use dynamic port list)	☒	0 ◆
	NonStop Process		
	Setting	Use Default	Value
	Source Process Name Starting With (Must start with "\$", followed by two letters, and end with "*". Example: \$AA*)	☒	
	Source Process CPU Number	☒	-1 ◆
	Source Process Priority	☒	0 ◆
	Target Process Name Starting With (Must start with "\$", followed by two letters, and end with "*". Example: \$AA*)	☒	
	Target Process CPU Number	☒	-1 ◆
	Target Process Priority	☒	0 ◆

Profile Settings

Category

- General
- Sorting Method
- Initial Compare
- Confirm-Out-Of-Sync
- Repair

Confirm-Out-Of-Sync

Setting	Use Default	Value
Perform Confirm Out-Of-Sync Step	☐	☒
Run Concurrently With Initial Compare	☐	☒
Delay Confirm-Out-Of-Sync By (seconds)	☒	0
Terminate when Maximum Records Out-Of-Sync	☒	100000
Output Out-Of-Sync Record Details to Report File	☐	☒
Update Report file Every (seconds)	☒	0
Update Report file Every (records)	☒	0
Source Oracle optimizer hint	☒	
Target Oracle optimizer hint	☒	

Event Reporting

Setting	Use Default	Value
Generate Messages	☒	None
Generate Warning Messages For Out-Of-Sync Rows After (differences)	☒	50

Agent

Setting	Use Default	Value
Use Static Listening Port For Agent During Row Hash On Source (0 to use dynamic port list)	☒	0
Use Static Listening Port For Agent During Row Hash On Target (0 to use dynamic port list)	☒	0

NonStop Process

Setting	Use Default	Value
Source Process Name Starting With (Must start with '\$', followed by two letters, and end with '*'. Example: \$AA*)	☒	
Source Process CPU Number	☒	-1
Source Process Priority	☒	0
Target Process Name Starting With (Must start with '\$', followed by two letters, and end with '*'. Example: \$AA*)	☒	
Target Process CPU Number	☒	-1
Target Process Priority	☒	0

Profile Settings

Category

- General
- Sorting Method
- Initial Compare
- Confirm-Out-Of-Sync
- Repair

Repair

Setting	Use Default	Value
Repair Batch Size	☒	10
Repair Transaction Size	☒	1
Number of Concurrent Repair Operations	☒	1
Check Changed Values	☒	☒
Terminate When Maximum Repair Warnings	☒	10000

Repair Reporting

Setting	Use Default	Value
Write Repair Success Messages to Report	☒	☒

Disable DB Triggers Session Based

Setting	Use Default	Value
Disable DB Triggers Session Based	☒	☐

Click 'Save' to save the changes.

2.9 Setup https on WebLogic and Veridata

Generate certificates for Load Balancer and Veridata server (ggpsppf501.sbg-psp-ppd.a.intuit.com for the following instruction)

The certificate for Load Balancer should include the domain name goldengate.sbg-psp-ppd.a.intuit.com, goldengate.sbg-psp-prod.a.intuit.com, goldengate-weblogic.sbg-psp-ppd.a.intuit.com, goldengate-weblogic.sbg-psp-prod.a.intuit.com. The file should include the following sections


```

-----BEGIN CERTIFICATE-----
...
-----END CERTIFICATE-----
-----BEGIN RSA PRIVATE KEY-----
...
-----END RSA PRIVATE KEY-----
Key Chain:
-----BEGIN CERTIFICATE-----
...
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
...
-----END CERTIFICATE-----

```

The certificate for Veridata server should include the following four files

```

ggpsppf501.sbg-psp-ppd.a.intuit.com.crt
ggpsppf501.sbg-psp-ppd.a.intuit.com.key
root.pem
intermediate.pem

```

- Login the Veridata server (ggpsppf501.sbg-psp-ppd.a.intuit.com) and sudo su - oracle.
- Perform the following steps to setup a keystore

Note: Need to replace <password> with the real password you choose

```

oracle@ggpsppf501 ~$ cd /u01/veridata/wls/user_projects/domains/base_domain/bin
oracle@ggpsppf501 bin$ . ./setDomainEnv.sh
oracle@ggpsppf501 base_domain$ cd /u01/veridata/wls/
oracle@ggpsppf501 wls$ mkdir keystores

```

Copy the certificate files to /u01/veridata/wls/keystores

```

oracle@ggpsppf501 keystores$ ls -ltr
total 16
-rw-r--r-- 1 oracle oinstall 2569 May 31 11:39 ggpsppf501.sbg-psp-ppd.a.intuit.com.crt
-rw-r--r-- 1 oracle oinstall 1492 May 31 11:40 root.pem
-rw-r--r-- 1 oracle oinstall 1787 May 31 11:43 intermediate.pem
-rw-r--r-- 1 oracle oinstall 1678 May 31 11:44 ggpsppf501.sbg-psp-ppd.a.intuit.com.key

```

```

oracle@ggpsppf501 keystores$ openssl pkcs12 -export -in ggpsppf501.sbg-psp-ppd.a.intuit.com.crt -inkey ggpsppf501.sbg-psp-ppd.a.intuit.com.key -out
server.p12 -name ggpsppf501.sbg-psp-ppd.a.intuit.com -CAfile root.pem -caname "Digicert CA Root"
Enter Export Password:
Verifying - Enter Export Password:

```

```

oracle@ggpsppf501 keystores$ keytool -importkeystore -deststorepass Intuit#m3 -destkeystore keystore.jks -srckeystore server.p12 -srcstoretype
PKCS12 -srcstorepass Intuit#m3
Entry for alias ggpsppf501.sbg-psp-ppd.a.intuit.com successfully imported.
Import command completed: 1 entries successfully imported, 0 entries failed or cancelled

```

```

oracle@ggpsppf501 keystores$ keytool -import -trustcacerts -alias intermediate -file intermediate.pem -keystore keystore.jks -storepass Intuit#m3
Certificate was added to keystore

```

```

oracle@ggpsppf501 keystores$ keytool -list -v -keystore keystore.jks -storepass Intuit#m3 > out.txt

```

```

oracle@ggpsppf501 keystores$ cat out.txt |egrep "Your keystore contains|Alias"
Your keystore contains 2 entries
Alias name: ggpsppf501.sbg-psp-ppd.a.intuit.com
Alias name: intermediate

```

```

oracle@ggpsppf501 keystores$ ls -l /u01/veridata/wls/keystores/keystore.jks
-rw-r--r-- 1 oracle oinstall 4317 May 31 13:17 /u01/veridata/wls/keystores/keystore.jks

```

- Add security group prod-veridata-vpc-3 as source with ports 7002 and 8831 to the security group prod-goldengate-vpc-3 through AWS console

- Configure WebLogic AdminServer to use keystone
 - You may check Configuring Oracle WebLogic Server (10.3.x - 12.1.x) to use SSL in Fusion Middleware 11g/12c (Doc ID 1235653.1) about the following steps
 - Login WebLogic Administration Console (<http://<dns name>/console/login/LoginForm.jsp>)
 - Select 'Environment' -> 'Servers' and click on AdminServer(admin)
 - Select the 'Keystores' tab
 - Select 'Keystore -> 'Change'
 - Select 'Custom Identity and Custom Trust' from the drop down list and click 'Save'
 - Enter the relevant information in the Keystores page:
 - 'Custom Identity Keystore' : /u01/veridata/wls/keystores/keystore.jks
 - 'Custom Identity Keystore Type' : JKS (Note: This has to be UPPERCASE)
 - 'Custom Identity Keystore Passphrase' : <storepass_pwd> e.g: welcome
 - 'Confirm Custom Identity Keystore Passphrase' : <storepass_pwd> e.g: welcome
 - 'Custom Trust Keystore' : /u01/veridata/wls/keystores/keystore.jks
 - 'Custom Trust Keystore Type' : JKS (Note: This has to be UPPERCASE)
 - 'Custom Trust Keystore Passphrase' : <storepass_pwd> e.g: welcome
 - 'Confirm Custom Trust Keystore Passphrase' : <storepass_pwd> e.g: welcome
 - Click 'Save'
 - Select the 'SSL' tab and enter the relevant information:
 - 'Private Key Alias' : ggpsppf501.sbg-psp-ppd.a.intuit.com (Veridata server name)
 - 'Private Key Password' : <keypass_pwd> e.g welcome
 - 'Confirm Private Key Password' : <keypass_pwd> e.g welcome
 - Click 'Save'
 - Select 'Environment' -> 'Servers' and click on the Managed Server configured
 - In the 'General' tab:
 - Check 'SSL Listen Port Enabled'
 - 'SSL Listen Port' : 7002
 - Click 'Save'

- Login to the Veridata server and then restart WebLogic

```
oracle@ggpsppf501 bin$ cd /u01/veridata/wls/user_projects/domains/base_domain/bin
oracle@ggpsppf501 bin$ . ./setDomainEnv.sh
oracle@ggpsppf501 base_domain$ cd /u01/veridata/wls/user_projects/domains/base_domain/bin
oracle@ggpsppf501 bin$ ./stopWebLogic.sh
oracle@ggpsppf501 bin$ nohup ./startWebLogic.sh > startWebLogic.log 2>&1 &
```

- Test https on Veridata server

```
oracle@ggpsppf501 ~$ curl https://ggpsppf501.sbg-psp-ppd.a.intuit.com:7002/console
<html><head><title>302 Moved Temporarily</title></head>
<body bgcolor="#FFFFFF">
<p>This document you requested has moved
temporarily.</p>
<p>It's now at <a href="https://ggpsppf501.sbg-psp-ppd.a.intuit.com:7002/console/">https://ggpsppf501.sbg-psp-ppd.a.intuit.com:7002/console/</a>.</p>
</body></html>
```

- Import the Load Balancer certificate using AWS Console
 - Go to Certificate Manager
 - Select Import Certificate
 - Click "Review and import". You should see the screen like below.
 - Click Import

- Change WebLogic Load Balancer prod-veridata-webloic configuration
 - Edit Listeners:

- Load Balancer Protocol: from HTTP to HTTPS
- Load Balancer Port: from 80 to 443
- Instance Protocol: from HTTP to HTTPS
- Instance Port: from 7001 to 7002
- SSL Certificate: <select imported certificate>

- Edit Health Check:

- Ping Protocol: from HTTP to HTTPS
- Ping Port: from 7001 to 7002

- Create a Route53 record set through AWS Console

- Name: goldengate-weblogic.sbg-qbo-prod.a.intuit.com
 - Type: CNAME
 - Value: prod-veridata-weblogic-498543478.us-west-2.elb.amazonaws.com (this is the Load Balancer prod-veridata's DNS)
- <https://goldengate-weblogic.sbg-psp-ppd.a.intuit.com/console> is WebLogic URL. Try it on browser
- Configure WebLogic Veridata to use keystone
 - You may check Configuring Oracle WebLogic Server (10.3.x - 12.1.x) to use SSL in Fusion Middleware 11g/12c (Doc ID 1235653.1) about the following steps
 - Login WebLogic Administration Console (<https://<dns name>/console/login/LoginForm.jsp>)
 - Select 'Environment' -> 'Servers' and click on VERIDATA_server1
 - Select the 'Keystores' tab
 - Select 'Keystore -> 'Change'
 - Select 'Custom Identity and Custom Trust' from the drop down list and click 'Save'
 - Enter the relevant information in the Keystores page:
 - 'Custom Identity Keystore' : /u01/veridata/wls/keystores/keystore.jks
 - 'Custom Identity Keystore Type' : JKS (Note: This has to be UPPERCASE)
 - 'Custom Identity Keystore Passphrase' : <storepass_pwd> e.g: welcome
 - 'Confirm Custom Identity Keystore Passphrase' : <storepass_pwd> e.g: welcome
 - 'Custom Trust Keystore' : /u01/veridata/wls/keystores/keystore.jks
 - 'Custom Trust Keystore Type' : JKS (Note: This has to be UPPERCASE)
 - 'Custom Trust Keystore Passphrase' : <storepass_pwd> e.g: welcome
 - 'Confirm Custom Trust Keystore Passphrase' : <storepass_pwd> e.g: welcome
 - Click 'Save'
 - Select the 'SSL' tab and enter the relevant information:
 - 'Private Key Alias' : ggqbpvd1pt.sbg-qbo-prod.a.intuit.com (Veridata server name)
 - 'Private Key Password' : <keypass_pwd> e.g welcome
 - 'Confirm Private Key Password' : <keypass_pwd> e.g welcome
 - Click 'Save'
 - Select 'Environment' -> 'Servers' and click on the Managed Server configured
 - In the 'General' tab:
 - Check 'SSL Listen Port Enabled'
 - 'SSL Listen Port' : 8831
 - Click 'Save'
- Login to the Veridata server and then restart WebLogic

```
oracle@ggpsppf501 bin$ cd /u01/veridata/wls/user_projects/domains/base_domain/veridata/bin
oracle@ggpsppf501 bin$ ./veridataServer.sh stop
oracle@ggpsppf501 bin$ nohup ./veridataServer.sh start >/dev/null 2>&1 &
```

- Test https on Veridata server

```
oracle@ggpsppf501 ~$ curl https://ggpsppf501.sbg-psp-ppd.a.intuit.com:8831/veridata/login.jsf
```

- Change WebLogic Load Balancer prod-veridata configuration
 - Edit Listeners:
 - Load Balancer Protocol: from HTTP to HTTPS
 - Load Balancer Port: from 80 to 443
 - Instance Protocol: from HTTP to HTTPS
 - Instance Port: from 8830 to 8831
 - SSL Certificate: <select imported certificate>
 - Edit Health Check:
 - Ping Protocol: from HTTP to HTTPS
 - Ping Port: from 8830 to 8831
- Create a Route53 record set through AWS Console
 - Name: goldengate.sbg-qbo-prod.a.intuit.com
 - Type: CNAME
 - Value: prod-veridata-2009728932.us-west-2.elb.amazonaws.com (this is the Load Balancer prod-veridata's DNS)
- <https://goldengate.sbg-psp-ppd.a.intuit.com/veridata/login.jsf> is Veridata URL. Try it on browser

3. Prepare for Veridata Agents Host in AWS

Follow steps in <https://wiki.intuit.com/display/EMSPD/Setup+veridata+agent>

4. Common Issues

1. Error on executing `java weblogic.WLST $temp_dir/create_data_source.py -p $temp_dir/myDomain-ds.properties`.

```
Starting an edit session ...
Started edit session, please be sure to save and activate your
changes once you are done.
This Exception occurred at Wed May 09 10:45:25 PDT 2018.
javax.management.InvalidAttributeValueException: Array has at least one null element
Problem invoking WLST - Traceback (innermost last):
  File "/tmp/setup_veridata_srv/create_data_source.py", line 92, in ?
    File "<iostream>", line 563, in set
    File "<iostream>", line 552, in raiseWLSTException
WLSTException: Error occurred while performing set : Error setting attribute Targets : Array has at least one
null element
Use dumpStack() to view the full stacktrace :
```

Workaround - In `create_data_source.py`, comment out the below two lines-

```
cd('/SystemResources/' + dsName)
set('Targets',jarray.array(\[ObjectName('com.bea:Name=' + dsTargetName + ',Type=' + dsTargetType)\],
ObjectName))
```

2. If the Target field is empty for all Data Sources in weblogic (Step 2.5a Verify Data Source), try adding the target (i.e, AdminServer, VERIDATA_Server_1) manually.
If veridata fails to start then stop the weblogic server, rename the \$MW_HOME (wls to wls.bak) and repeat the steps from 2.2.

3. There could be issue while running command "`java weblogic.WLST $temp_dir/create_data_source.py -p $temp_dir/myDomain-ds.properties.opss`"
Solution : Set CLASSPATH env variable as below
`export CLASSPATH=$CLASSPATH:$JAVA_HOME/lib/tools.jar:$MW_HOME/wlserver/modules/features/wlst.wls.classpath.jar`

4. There could be issue where admin server listening to 127.0.0.1. Change /etc/hosts as below:

```
[oracle@ggpspiob1 scripts]$ cat /etc/hosts
# Your system has configured 'manage_etc_hosts' as True.
# As a result, if you wish for changes to this file to persist
# then you will need to either
# a.) make changes to the master file in /etc/cloud/templates/hosts.redhat.tmpl
# b.) change or remove the value of 'manage_etc_hosts' in
#    /etc/cloud/cloud.cfg or cloud-config from user-data
#
# The following lines are desirable for IPv4 capable hosts
#127.0.0.1 ggpspiob1.sbg-ppd.a.intuit.com ggpspiob1
127.0.0.1 localhost.localdomain localhost
127.0.0.1 localhost4.localdomain4 localhost4

# The following lines are desirable for IPv6 capable hosts
#::1 ggpspiob1.sbg-ppd.a.intuit.com ggpspiob1
::1 localhost.localdomain localhost
::1 localhost6.localdomain6 localhost6
```